A Survey on Organic Agro Data Towards Agriculture Using Data Mining

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***Abstract*— In India, 60.6% of the land has been used for agricultural purposes. Farmer invests much in chemical fertilizers, manual labor, weed collection, pesticide management, and at last, they gain low productivity. In this paper, different data mining techniques are going to be analyzed for improving crop productivity and the use of organic farming. Here nutrient content of different food products is also going to be analyzed by comparing organic and inorganic food products. Ill effects of Inorganic farming are listed. Food produced using organic farming will bring out the health and well-being of human beings for a better future. Seasonal crop recommendations will also be done by making a complete study on an attribute such as PH, water holding capacity, erosion, etc. Weather data are collected to bring out the prediction of rainfall in a particular region. The accuracy of different data mining algorithm are also listed**

Keywords*— *Organic farming, Inorganic farming, Data Mining, Agriculture, Technology

I. INTRODUCTION

Data Mining is the progression of discovering interesting patterns and knowledge mining[35] from huge data. The input data are collected through autonomous flight, satellite images, online domain, agriculture monitoring etc which is shown in fig. 1. Here various data mining technique is analyzed for different agriculture problem such as data for different soil which are suitable for different food products, the importance of organic fertilizers, harmful due to Inorganic fertilizers, different climatic condition to grow different food products, and Finally, nutrient content of food products is compared with organic and inorganic food products is compared. Proper training needs farmers on organic farming to update them on organic farming. Hence a lot of methods and principles are discussed in this review to illustrate the pros and cons.

II. ORGANIC AGRICULTURE AND DATA MINING METHOD

Organic agriculture is a farming method in which natural fertilizers and pesticides are used for the well being of humans. The method used in organic agriculture are green technology method, crop rotation, cover cropping, reduce tillage, compost application. The different organic agriculture and data mining methods are highlighted in following.

A. Different soil is classified with their attributes like water retention, air circulation, and mineral content as proposed by N.Hemageetha[1](2016) et. al . Soil is classified into three (a) sandy which has high air circulation, medium mineral content, and poor water retention (b) clayey which has poor air circulation, medium mineral content and lofty water withholding (c) loam which has high air circulation, high mineral content and towering water preservation. Applying the data mining techniques to the collected samples using the weka tool. As a result benefits of understanding, the soil has improved the productivity of crops, and reduced fertilizers.

B. Valmik B Nikam[2](2013) et. al. has projected a forecast sculpt for rainfall. Weather information from the metrological department is collected and it helps predictrainfall. Collected data consist of 36 attributes, out of which 7 attributes are used for the prediction of rainfall. Weather forecast is divided into the following categories (i) Now forecast which is used to forecast for a few hours (ii) Short- range forecasts is used to forecast for 1 to 3 Days (iii)Medium-range forecast is used to forecast for 4 to 10 days (iv) Long range forecast is used to forecast for more than 10 days. Weather forecasting and research models, synthetic forecasting models, Seasonal climate forecasting, and universal data forecasting models are currently used as a model that is considered expensive. Using data mining author has created an inexpensive model for rainfall prediction. By applying the Bayesian rainfall prediction model, using the sample data the rainfall prediction is opted out. The accuracy may be degraded since only 7 attributes are considered.

C. An automated pest detection and classification in the fruit crop was proposed by Agustina Suarez[3](2021) et. al. The input images are collected from the field. Deep neural-based classification was used for image processing. Finally, An accuracy of 94.8% was obtained. However the system used only limited dataset.

D. A crop selection method has been proposed by S.Pudumalar[4](2016) et. al. Farmer can choose the rightcrop for their field by analyzing soil requirements. If the right crop is not chosen then there will be a serious setback in crop productivity. Precision agriculture is site-specific farming which helps provide a better decision on farming. A crop selection method where a series of input is going to be given by considering various factors like weather, soil type[34], crop variety, and water density. The predicted rate determines the accuracy of CSM. Here various attribute is depth, texture, PH, soil shade, permeability, drainage, and water withstand ability. These attributes help in the ability to exact water and nutrient from the soil. Various data mining techniques like Random tree, CHAID, K-Nearest and Naive Bayes are used. The simple rule-based was also demonstrated for illustration purposes. However data size is found to be less and prediction of yield is not addressed much.

E. A crop row detection in organic agriculture was developed by Vitali Czymmek[5](2020) et. al. using a semi-autonomous flight. Input to the system is a series of field images. A qualitative assessment was performed with the input. As a result, organic farming is possible with smart farming was studied. However the study is conducted only for plant inner row, Intra row is not considered.

F. An automatic tomato mature estimator was developed byPrasenjit Das[6](2020) et. al. The input data are collected through an online domain which consists of 50,000 images of fruits and vegetables. Using AlexNet framework, classification was done through a convolution neural network. As a result, 8 classes of distinct tomato images are classified. The proposed system has a limitation that it was not trained for real time tomato classification.

III. ORGANIC AGRICULTURE AND ITS APPROACH

Organic agriculture is a ecological farming that uses fertilizers of organic origin. Many approaches are highlighted in the following section for improving the crop yield of organic agriculture.

A. A government-supported scheme for organic farming has been proposed by Ke Nunthasen[7](2011) et. al. Organic agriculture was promoted by the Thailand government by conducting different programs and also three different trends have been discussed which increased the popularity of organic agriculture (i) Public awareness for healthy living (ii) Sustainable agriculture development (iii) Rise of environmental awareness. Due to the step taken by the Thailand government the entire area under organic farming has improved dramatically. The overall area beneath organic farming in 1998 is 6,281 rais whereas in 2009 it is 1,84,981 rais. In 1998 & 1999 only field crops were planted. In 2001 & 2002 field crops and vegetable were planted from 2004 to 2009 all the possible cultivation were done using the organic farming. The total area under organic cultivation has increased from 6281 to 184981 rais. The government has also conducted various training agriculture programs to improve organic agriculture. The various programs are as follows (i) TAPTMS (Agriculture program in trade and marketing (ii) TIE (Integrated education) (iii) TNRENV (Natural resource and environment conservation) (iv) TEK (Knowledge of technology)

B. S.Selvi[9](2012) et. al. has proposed technology for environment-friendly agriculture. By using modern agriculture high variety of seeds, compound fertilizers, water, pesticide, etc are used to gratify the food stipulate. Due to this modern agriculture, natural resources are exhausted. An alternative type of agriculture practice organic agriculture is done to compensate for the ill effect ofconventional farming. Organic farming not only increases productivity but also gives healthy food crops with balancedvitamins and minerals. Estimates by the Asian Development Bank (ADB) show that land use per person will decrease from 0.17 hectares in 1990 to 0.12 hectares in 2010. However organic farming is practiced in more than 130countries all over the world up to 15 % of total farming. Organic farming produces eco-friendly. Countries such as Japan, EU, USA, India, Brazil, Argentina and Switzerland have adopted organic standards and established inspection and certification mechanisms. Uttarakhand, Mizoram, Sikkim, Karnataka, Maharashtra, Tamilnadu, and Kerala arethe states that have adopted policies and programs on organic farming. India produces a broad variety of organic products from fruits, vegetables, spices, food grains, pulses, milk, and organic cotton. SWOT analysis is a tool used to analyze the overall strategic position of a business and industrial sector. Here SWOT is used to analyze organic farming and its perspectives.

C. Mahesh PJ[10](2016) et. al. have proposed new technologies for cultivating organic farming. The different methods are (i) Hydroponics, which is a technique where less soil is used and PVC pipes are implanted for growing crops (ii) Aquaponics, in this technique organic crops are planted without using soil in a special grown bed (iii) Aeroponics is a technique in which the mist is used as a medium to grow plants.

D. T.J LaSalle[11](2008) et. al. has illustrated the ill effect of modern agriculture that cause global warming at a greater rate. Rodale Institute has compared inorganic and organic farming systems. The modern farming system breaks down the soil carbon into carbon dioxide which contributes to global warming to a greater extent. Using synthesis fertilizer will degrade the soil which also affects human and animal health. There is research says that the US farm system contributes to 20% of the nation's carbon dioxide emissions. However other toxic gas were not considered that affect the plant growth Using the regenerative organic farming method we can remove carbon sequester from the soil in the air.

E. Rajesh Kumar Dubey[12] (2015)et. al. have proposed a green technology method using organic agriculture. In modern agriculture not only chemical is added to farming land but also to the water and food we consume. Organic farming increases soil fertility and crop nutrition is maintained. Soil fertility, food security, and nutrition security are discussed briefly. Food security means that all people around the globe get quality food with all nutrition in it. Nutrition security is the food must maintain quality vitamins in it. The World summit in 1996 brings focused on food, health, and nutrient. Time to bring out a green revolution in an eco-friendly way so that food security and nutrient is maintained.

F. Rajesh Kr Dubey[13](2014) et. al. have proposed eco-friendly technology for organic farming. Various problems of inorganic farming are discussed such as water logging, a decline in groundwater level, the use of different pesticides that kill all micro organisms & groundwater contamination. Organic farming crops have better nutrients and high protein value than conventional farming crops. The government of Sikkim, Mizoram, and Uttarakhand have taken so many steps to make their states complete organic farming agriculture.

G. Husnawati[14](2013) et. al. have proposed deep research into Taiwan rice production. As the years go there is ademand for organic rice in Taiwan. Organic farming was 159.6 in 1996 and 2003 it was 1092 hectares. Here is a newtechnique called rotational irrigation where water is suppliedto crops at an ideal time. There are different types of rice varieties (i)Taichung (ii)Tainung (iii)Kaohsiong (iv)Tainanare farmed in Taiwan. The Taiwan government is providing organic certification to farmers after inspection onsite. This certification is valid only for 3 years and it must be renewed after that. For attaining this certificate farmer's crop must be free from chemical fertilizers. This study concentrates only on rice production however other crops can also be considered.

H. Sohel Rana[15](2017) et. al has made a complete study on farmer attitudes toward organic vegetable farming. In this research, 95.4% of farmers have a positive attitude. Data was collected during direct individual interviews with farmers. A set of predefined questionnaires have been framed to collect the data in 2015 in Bangladesh and also farmers have produced 3729 thousand metric tons of vegetables in 2015 - 2016 on 989 thousand acres. No practical demonstration was not done.

I. Tomek de Ponti[18](2012) et. al. has proposed a yield gap between organic and conventional agriculture. Issues faced by organic farming are(i)organic farming will be able to compete with conventional farming since the farming area islesser than the organic farming (ii) The cost of organic farming products is higher (iii) Organic farming requires more farming land (iv)food security. The yield between organic and inorganic farming depends upon climate, cultivating area, use of organic fertilizers, and crop that can sustain pests and other diseases. Here data are analyzed for5 crops wheat, corn, barley, corn, and soya bean. The yield of soybean is 80% higher than the other crops whereas potatoes score lower than 80%. More numbers of crop canbe considered for organic crop growth.

J. E.Joanna Kezia[19](2013) et. al. conducted a field experiment with the radish plant. The effect of organic and inorganic fertilizer was analyzed. The parameter measured for the study is weight, leaves, and length of the bulb. Inorganic fertilizer had a noteworthy blow on weight, leave but not on the length of the bulb on the radish plant. Four unique combinations of organic and inorganic fertilizer were applied to the field. Organic fertilizer combinations are field soil and vermin compost. Inorganic fertilizer combinations are super phosphate and potash. The study reveals that there is no difference in weight and leaves of radish plants where fertilizer is affected only by the length of the radish plant. Only limited data was considered.

K. Xiao-Juan Liu[20](2012) et. al. has proposed yard waste that is used for organic mulch. This study will bring out the effect of yard waste on soil fertility. Mulches are used to maintain soil temperature, weed growth, and retain soil moisture. This study brings about the preparation of organic mulch with experiment treatment. The organic matter in the soil is decomposed as any form of nitrogen that is taken by plants for growth.

L. Thomas F.Doring[21](2014) et. al. have done brief research on organic farming. Over the years there is an increase in organic farming. Hence it is necessary to conduct a wide range of different research on organic farming. This farming system has brought out the solution to different problems like climate change, food security, and biodiversity conservation. This paper solves the different needs for solving organic farming that is challenging to the universe.

M. Jigme[22](2015) et. al. has conducted a brief study on the consequence of organic and inorganic fertilizer on broccoli plant. For this purpose, 3 different types of organic fertilizers and 2 types of inorganic fertilizer are chosen. For organic fertilizer treatment chicken manure tea(CMT) is used. The parameter measured is stem diameter, leaf number, leaf diameter, plant weight, and height. Vegetable growth was analyzed from a randomly used 10 samples. It has been observed that nutrient content & vegetable growth of organic fertilizer crop is much higher than an inorganic broccoli plant. Hence a significant plant growth was seen on inorganic fertilizer and 200 ml/week of CMT organic fertilizer. Only less number of samples are considered.

N. Madhusudhan[23](2016) et. al. has proposed an eco-friendly way of agriculture by using organic farming. Agriculture has become a high investing sector since the useof synthetic chemicals whereas farmers get low yield and it also affects the ecosystem. In organic farming, synthetic fertilizers are replaced by bio pesticides, bio fertilizers. Farmers use synthetic fertilizers to protect plants from insects and pests and also give high yields. As the day goes by pesticides and insects get resistant to that synthetic fertilizers. Therefore these fertilizers become useless at one stage. Many countries import and export fruits, vegetables, seeds, etc some of the countries have banned imports since the use of high chemical content. Organic farming has various techniques like crop rotation, vermicomposite, animal husbandry, bio fertilizer, and green manures.

O. Catherine Badgley[25](2007) et. al. has given a model for organic food supply. Here the organic and inorganic food production are compared with 293 samples. The average yield is <1.0 in developed countries and >1.0 in developing countries. In the global food production method, FAO(Food and Agriculture Organization) has analyzed whether a country is a developed or developing country with the help of yield ratio. Two models are followed (i) Comparison of organic and inorganic food production and (ii) Food production yield ratio all over the world.

P. S.Hazarika[28](2013) et. al. have proposed an organic farming area under cultivation, and food security and food organization in India has been discussed. There is a list of accredited inspection and certifying agencies in India. They are (i) Association for promotion of organic farming (ii) Indian organic certification agency (iii) Indian society for certification of organic products etc. There are also some international standards (i) IFOAM (ii) CODEX (iii) JAS (iv) EU regulation. To improve the soil fertility LTFE(Long Term Fertility Experiment) was conducted to improve the organic farming soil fertility. To improve soil fertility animal manure is used in organic farming.

Q. MA. Sarker[29](2011) et. al. studied organic wheat production in Bangladesh. Bangladesh farmers produce a wide variety of wheat that is not prone to insects and pests. These wheat are exported all over the world. The different types of wheat varieties that are produced are Shatabdi, Sufi, Bijoy, and Prodip. Cattle dung is the major fertilizer which was 80 million tons in 2000. Crop duration of BARI wheat 21 & BARI wheat 22 is 105 days, BARI wheat 23 is 103 days, and BARI wheat 24 is 102 days. The average yield production in 2000 is 876 acres. The major limitaion is that only single crop was considered.

IV. ORGANIC AGRICULTURE COMPARATIVE AND CASESTUDY *A.*

In this section, organic agriculture comparative and casestudy are highlighted.

A. E.Thippeswamy[8](2013) et. al. has proposed a brief comparison between the inorganic and organic food products. Extensive use of chemical fertilizers in inorganic farming causes harm to human beings and also it spoils the agricultural land. Due to inorganic farming humans have problems that are related to heart attacks, strokes, cancer, and many other diseases. Given the safety of living creatures organic type of farming is being promoted. The organic method of farming taste better and also contain better balance vitamins and minerals than inorganic food. Inorganic has added 130 diverse classes of pesticides containing additional 800 items. Pesticides enter the food chain in 4 ways (a) pesticide for on-farm use (b) pesticides for post-harvest (c) imported food pesticide use (d) canceled pesticides. Whereas organic type of farming will bring out a natural way of food products that will not cause any harm tothe human being.

B. Vishnu shanker Meena[16](2015) et. al. have proposed a different recent advancement of organic farming in the state of Uttarakhand. The secondary data collected from UOCB(Uttarakhand organic commodity board) is used for analysis. The government has proposed organic farming methods where traditional farming like green manure, biological pest management, etc. The organic farming area in India is 42,000 hectares in 2004 - 05 and 1.08 hectares in 2010. In Uttarakhand, 8% of farmers are practicing organic farming through the rain-fed condition. UOCB has taken measures to promote organic agriculture in Uttarakhand.

C. Raj Kumar Banjara[17](2016) et. al. has made a complete study on organic farming in Nepal. This study wasconducted at Nepal in 4 districts with 614 organic farmers. It has brought the social status of farmers improved and alsothere is an increase in farmers with environmental factors. This study brings out the improvement that is needed in infrastructure development, policy development, and motivation of farmers. Government must manage three main sectors of organic agriculture (i) Promotion and knowledge of organic farming (ii) Availability of seed and fertilizers (iii) Marketing strategy. The Nepal government has formed the Nepal Organic Agriculture Development Board(NOADB) for the development of organic farming. Finally, it has been found that 90% of the farmers are satisfied with their income in organic farming.

D. Nasreen Anjum[24](2017) et. al. has proposed an NPOP(National Program on Organic Production), targeted sample organic producers to study the knowledge of organic dairy farming. This sample was collected from 10 farmers of four villages. Totally 120 dairy farmers were selected for the sampling method. This sampling method contains the analysis of health, nutrient, marketing, certification, and production. As a result, organic dairy contains good taste food with no chemical content in it. Significant health benefits are also observed but limited samples are taken.

E. A comparative study for organic agriculture has been proposed by Kritika Gurung[26](2013) et. al. The organic agriculture in India is compared with all other leading organic agriculture countries in the world. Australia leads the way with 12 million hectares of farmland while India has 1.18 million hectares of farmland. Europe takes 28% of organic farming land in 2010. There are 3 organic regulatory(i)Legal instruments has worked with Switzerland, Norway, etc, (ii) Financial instruments - Many countries provide financial support (iii) Communicative instruments provide awareness of organic farming. United state of America has a regulatory USDA(United states department of agriculture) for organic agriculture development.

F. Raghavendra kondaguri[27](2014) et. al. has proposed a complete study on organic and inorganic paddy yield. In this different types of input are used for paddy cultivation. Eight types of input are used for organic paddy and seven types of input are used for inorganic paddy. Organic paddy uses high manual labor and inorganic paddy uses low manual labor. sixty samples are collected from organic and inorganic farming paddy. As a result, farmers use very less seed in organic paddy where the germination percentage is much higher so the cost involved becomes less. The yield of organic paddy is low since the farmers are practicing organic farming for very few years. So it takes time to build organic farming. Hence yield will be high after a few years of organic farming.

The outline of data and method is shown in Table.1 and Contribution to organic agriculture using data mining, method and approach is shown in Fig. 2.

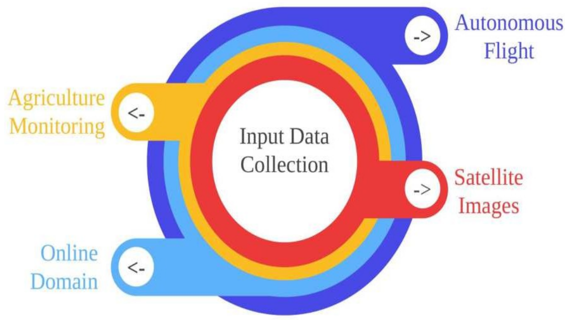

Fig. 1. Input data collection

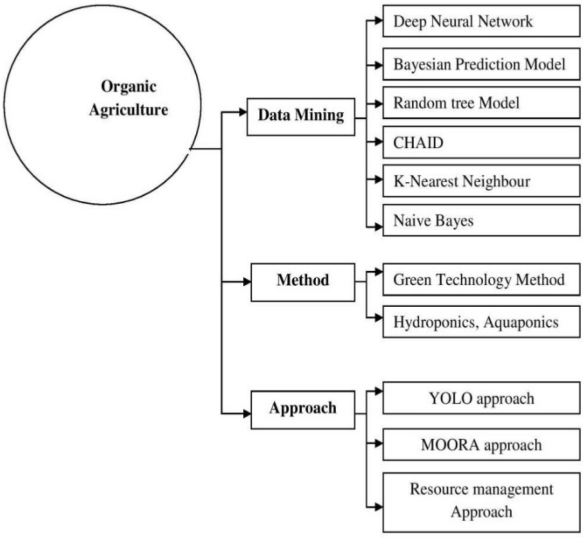

Fig. 2. Contribution To Organic Agriculture Using Data Mining, Method And Approach

TABLE I. OUTLINE OF DATA & METHOD

S. No	Author	Source of Data	Data Acquisition equipment	Processing Methods/ Approach	Purpose
1	Hasnelly[30] (2011) et. al.	Respondents of Organic product industry	Individual	Resource management Approach	Customer value and satisfaction
2	Vitali Czymmek[31] (2019)et. al	Live farm data	Multi platform robot	YOLO approach	Weed detection in Carrot field
3	Daniel Caravantes[32] (2021) et.al	APBOSMAM farm data	Monitoring	Segmentation algroithm	Develop and monitor organic banana
4	Aishwarya K S[33] (2018) et. al.	Sensor	GSM module	Aquaponics Approach	Automated feed for fish and plant

V. CONCLUSION

Using different data mining techniques in organic agriculture can improve crop productivity. Using soil database, rainfall prediction, and crop recommendation techniques we can have a better agriculture process and better crop production. A comparison is made between organic and inorganic food which helps people to have better health. As a suggestion, Organic crop recommendations will help the farmers to increase their production. Organic and inorganic crop production, food security, ecosystem, and environmental changes in different countries are also studied. The different data mining algorithm was studied and limitation each are also discussed. As a future scope, Organic agriculture with technology integration will help in better organic crop productivity and also organic production can be studied with different crop since only limited crop like rice, radish, soybean are considered. The nutrient content of organic crop will also be studied to illustrate the benefit to end user.

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